CSCI 4150 · INTRO TO AI · A2 REVIEW

Spending Analyzer

Sorts a student's bank charges into categories and suggests a budget

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SEPTEMBER 2026 · CARD ••4821

SQ *KOPI KOPI TROY NY	6.18
TARGET T-1849 LATHAM NY	28.41
UBER *EATS HELP.UBER.COM CA	22.68
PLANET FIT CLUB FEES	15.00
STEWARTS SHOP 345 TROY NY	28.81
DOORDASH*DASHPASS	4.99
VENMO PAYMENT 5225551094	29.43
PAYPAL *KJ8STUDIO	18.00
AMAZON PRIME*DJP7WVGSQ	7.49
DD *DOORDASH SUSHIKING	25.95
SPENT IN SEPTEMBER	**2,053.54**

To budget, a student first has to figure out what `SQ *KOPI KOPI` is

WHO IT'S FOR

Students living off campus, paying rent, bills and subscriptions for the first time on a part-time paycheck.

WHAT IT HAS TO DO

1. Name the category of every charge
2. Find the recurring bills
3. Suggest a budget they can keep

WHAT IT CAN'T GET WRONG

Every dollar total has to be exact.

On the statement	What it was
`SQ *KOPI KOPI TROY NY`	Coffee shop
`PLANET FIT CLUB FEES`	Gym membership
`DOORDASH*DASHPASS`	Delivery subscription
`STEWARTS SHOP 345`	Gas or ice cream
`TARGET T-1849`	Food or home goods

For mixed stores like the last two, the same text can mean different purchases.

Code does the math. An LLM sorts charges and asks the student when unsure.

INPUT
Bank CSV
date, text, amount

→

CODE
Clean up
strip codes and store numbers

→

STUDENT + LLM
Categorize
1 Reuse past answers if they agree
2 LLM categorizes the rest
3 If unsure, ask and save the answer

→

CODE
Compute
totals, bills, budget math

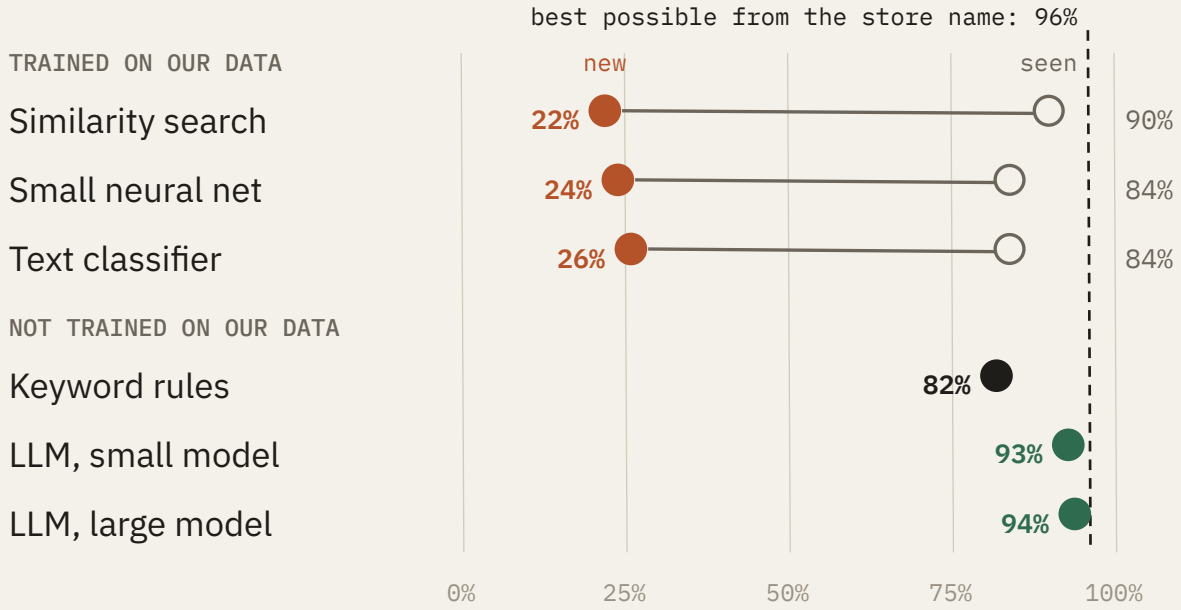
→

LLM
Explain
sums up our numbers in words

WHERE THE MEANING COMES FROM

the bank's charge text	regex patterns we wrote	our 11 categories, what the LLM knows about merchants, and the student's answers	exact sums, dates, monthly repeats	a prompt with only our numbers

Our trained models fell from about 85% to 25% on merchants they hadn't seen



They memorized merchant names. On a new one like NETFLIX, the classifier guessed Transport.

The LLM knew most merchants. Every miss was at a mixed store (Target, Costco, Stewart's, Apple).

Accuracy = picked the same category we did by hand. Trained models: 5-fold cross-validation over 211 transactions. “New” = the store was never in training. Dashed line: 3 charges share a store name with a different purchase.

On new merchants, the classifier's most confident answers were all wrong

New merchant	Classifier said	Confidence	Actually
UBER *EATS	Transport	0.91	Dining
AMAZON PRIME*	Shopping	0.91	Subscriptions
DOORDASH*DASHPASS	Dining	0.86	Subscriptions
UBER *TRIP	Dining	0.85	Transport
TROY PARKING VIOLATION	Transport	0.81	Fees

0 of 28

confident guesses (above 0.70) were right, over all 3 months

Similarity search did the same: UBER *EATS and UBER *TRIP score 0.80.

The scores from our trained models can't tell us when to ask the student. We need a signal that goes up when the text is actually unclear.

The LLM got 63 of 68 categories right and matched our total to the cent

Condition	Result
No statement	Didn't guess: “I can't see your spending from here.”
Statement, small model ×3	**63 of 68** right, identical in all 3 runs. Total **$2,053.54**, same as our code.
Statement, large model	**64 of 68**, one short of the limit. All 4 misses marked unsure.
Our numbers only	Stuck to our numbers and asked about the guesses we flagged.

LLM OUTPUT, RUN 1 (EXCERPT)

```
22 | Dining        | sure    LITTLE MOMO
39 | Health        | sure    PLANET FIT
26 | Subscriptions | sure    DASHPASS
14 | Groceries     | unsure  Stewart's: gas
18 | Shopping      | unsure  Target: food
66 | Groceries     | unsure  Costco: chair
```

When run 1 said sure: **58 of 59** right

When it said unsure: **5 of 9** right

Every row uses Claude Haiku 5.5 except the large model (Claude Opus 5.5). No tools. Grey notes are ours.

Our evidence supports code for the math and an LLM for new merchants

ADOPT

Code for all totals
Always exact, used to check the LLM's total

LLM summary from computed numbers
Used only our numbers (1 run so far)

MODIFY

LLM categorizer + ask when unsure
93–94% right, 17 of 19 misses flagged (4 runs)

Recurring-bill rule (utilities vary)
Found 11 of 13, missed electric ($38 → $61)

REJECT

Keyword rules for categories
82% vs. the LLM's 93%, gym filed as Fees

Our classifier and similarity search
26% and 22% on new merchants

DEFER

Small neural net
No better than the classifier (24% vs. 26%)

Receipt scanning (OCR)
Could settle mixed stores (12% of charges)

Can a student's own past answers settle Target, Costco and Stewart's?

WHY THIS QUESTION

12% of charges are at stores that sell several kinds of things. In our first run, they were 7 of the LLM's 9 unsure flags.

WHAT WE'LL TRY

Before asking, give the LLM the student's confirmed past charges at that store, with amounts (a simple form of RAG).

HOW WE'LL JUDGE IT

Questions per month and charges still wrong, compared to always asking.

STEWARTS SHOP 345 · WHAT THE LLM WOULD SEE

Jul 03	$25.64	gas	confirmed
Jul 25	$5.63	ice cream	confirmed
Aug 08	$5.67	ice cream	confirmed
Aug 26	$33.59	gas	confirmed
Sep 06	$28.81	?	
Sep 24	$5.53	?	

Hypothesis: with this history, big means gas and small means ice cream. We haven't tested it yet.

How we tested

TEST SET

211 transactions for one student, Jul–Sep 2026: 66 merchants, 11 categories. Synthetic, written in real bank and card statement formats, and labeled by us. Mixed-store charges are labeled by what was bought.

APPROACHES

Keyword rules: 112 brand and word patterns. Classifier: TF-IDF character n-grams + logistic regression. Neural net: MLP, 64 hidden units, same features. Similarity: nearest labeled charge by cosine.

EVALUATION

5-fold cross-validation over all 211 charges. “Seen”: random folds. “New”: each merchant's charges held out together. Slides report the 68 September charges, the same ones the LLM saw.

LLM TEST

The 68-line September statement plus our 11 category names. Claude Haiku 5.5, 3 runs. Claude Opus 5.5, 1 run. No tools. Also a no-statement run and a numbers-only run.

LIMITS

One synthetic student and our own labels. On 68 charges, one charge moves accuracy 1.5 points.

Evidence summary

Approach tested	Hypothesis	Evidence (Sept, 68 charges)	Decision	Why
Keyword rules	Store names imply category	82%, gym charges filed as Fees	Reject	LLM: 93% on the same charges
Text classifier	Learns from our examples	84% seen, 26% new merchants	Reject	Confident when wrong
Small neural net	More capacity, better fit	84% seen, 24% new merchants	Defer	Needs many students' data
Similarity search	Similar text, same category	90% seen, 22% new merchants	Reject	Similar text ≠ same category
LLM categorizer	Knows unseen merchants	93–94%, 17 of 19 misses flagged	Modify	Ask the student when unsure
Code for totals	Totals must be exact	Exact, LLM matched it 4 of 4 runs	Adopt	Code is the answer key
Recurring rule	Same amount each month	11 of 13, missed electric, iCloud	Modify	Let utility amounts vary
LLM explanation	Explains, adds no numbers	Only our numbers, 1 run so far	Adopt	Numbers stay traceable
Receipt OCR	Receipts show the items	Untested, 12% of charges affected	Defer	Try past answers first

Accuracy: the 68 September charges the LLM saw. Bill check and 12%: all 3 months. “New” = store never in training.

TEAM CONTRIBUTION SNAPSHOT

Who did what:

https://drive.google.com/drive/folders/1ON98bTv1s4N0X8jsCrmlbD7CLP3OIScL?usp=sharing

Team member	Main responsibility in A2
Josh Ramirez	System Map, Training Models
Neil Paul	Brainstorming Approaches, System Design
Neil Shrestha	Test data, All experiments, Evidence table, Slides